Assignment 7, explained

Abalone age groups: logistic regression, PCA, and SVM

A plain-language walkthrough of both parts,
with the numbers from the solution notebook.

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What the file is, and why Rings becomes a class

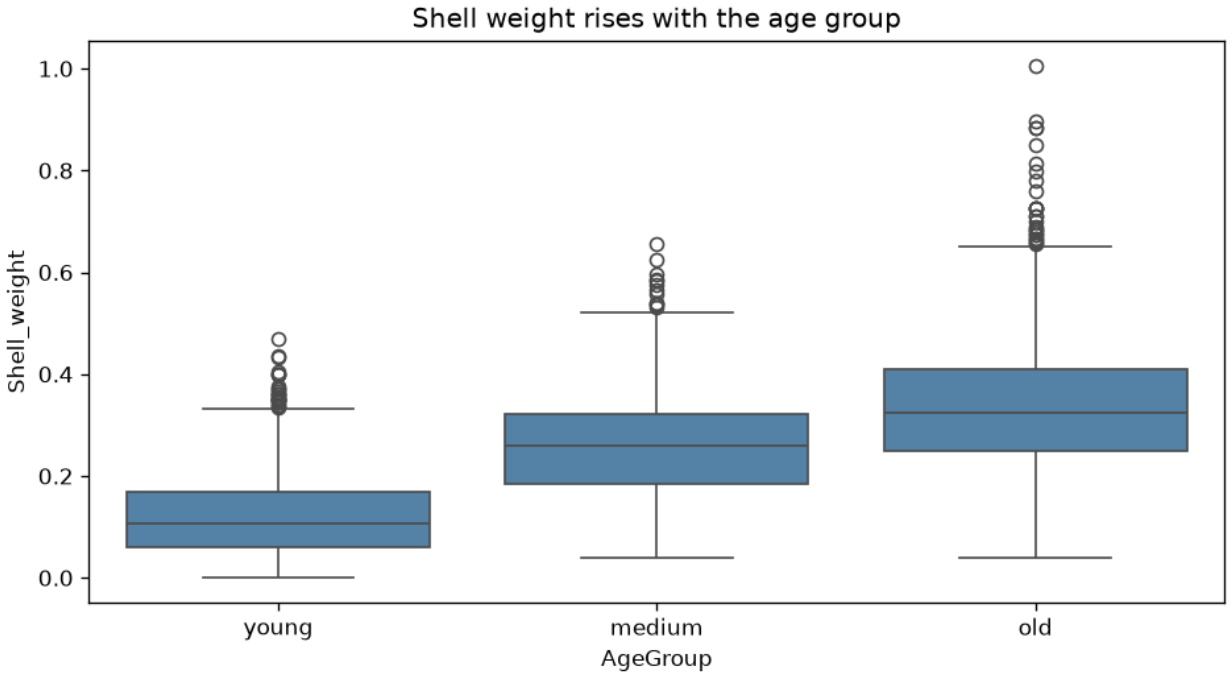
UCI Abalone has 4,177 animals and no header. The columns, in order, are Sex (M, F, or I for infant), Length, Diameter, Height, Whole weight, Shucked weight, Viscera weight, Shell weight, and Rings.

Rings is a count. Adding 1.5 is a rough age in years. Logistic regression and SVM, as used in this lab, predict a category, so the count is cut into three groups of similar size before any model is trained:

- young: 8 rings or fewer (1,407 animals)
- medium: 9 or 10 rings (1,323)
- old: 11 rings or more (1,447)

Those cuts sit near the quartiles of this file. Rings is not given to the model as an input. If it were, the label would be a copy of a feature and the accuracy would be meaningless.

Sex is text. One-hot encoding makes dummy columns. One of the three is dropped, because M + F + I would always equal 1 and would hand PCA a column with no information. The remaining inputs are then standardized: zero mean, unit variance, using the training rows only.



Older groups have heavier shells. Size and age move together, which is why the first principal component will be 'overall size'.

Say this: I turned Rings into young / medium / old, and I did not feed Rings back in as a feature.

Part A. Logistic regression on the scaled features

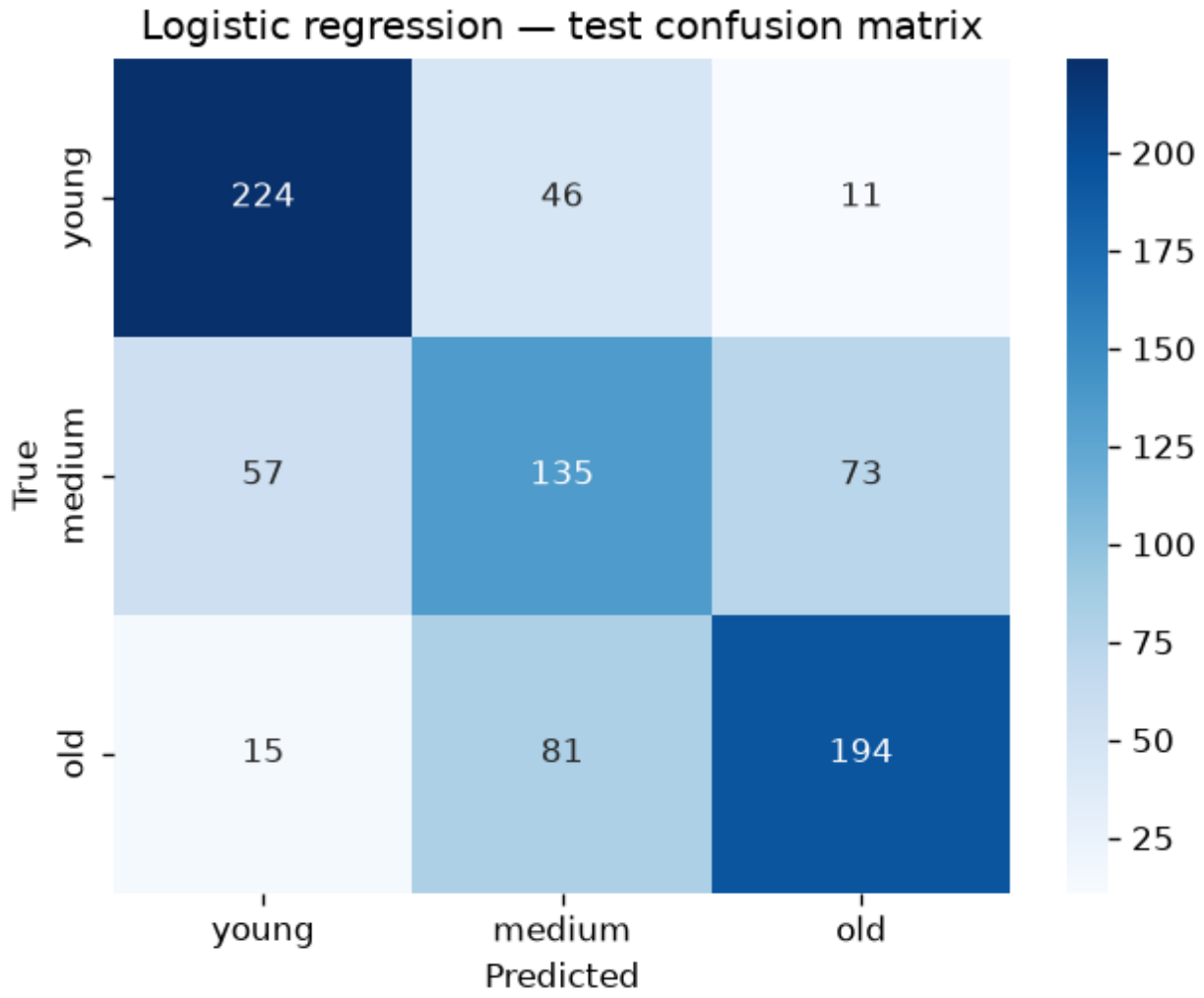
Same split for every model later: 80/20, stratified, random_state=42. That is 3,341 train and 836 test. The scaled matrix has 9 columns: 7 measurements plus Sex_I and Sex_M (female is the dropped baseline).

With three classes, logistic regression learns one score per class and softmax turns the scores into probabilities that add to 1. Current sklearn does this automatically. The old multi_class='multinomial' argument has been removed.

Test accuracy is 66.15%. Young and old are easier. Medium is the overlap: precision and recall are both about 0.51. A medium animal is often called young or old, which is expected because 9–10

rings sits between the other two groups.

True class	Precision	Recall	F1	Support
young	0.757	0.797	0.776	281
medium	0.515	0.509	0.512	265
old	0.698	0.669	0.683	290



Most mistakes are one group away. Young is rarely called old, and the other way around.

Part A. PCA, then logistic regression again

PCA rotates the 9 standardized columns into new axes that are uncorrelated. The first axis is the direction where the animals differ most. explained_variance_ratio_ is the share of variance on each axis:

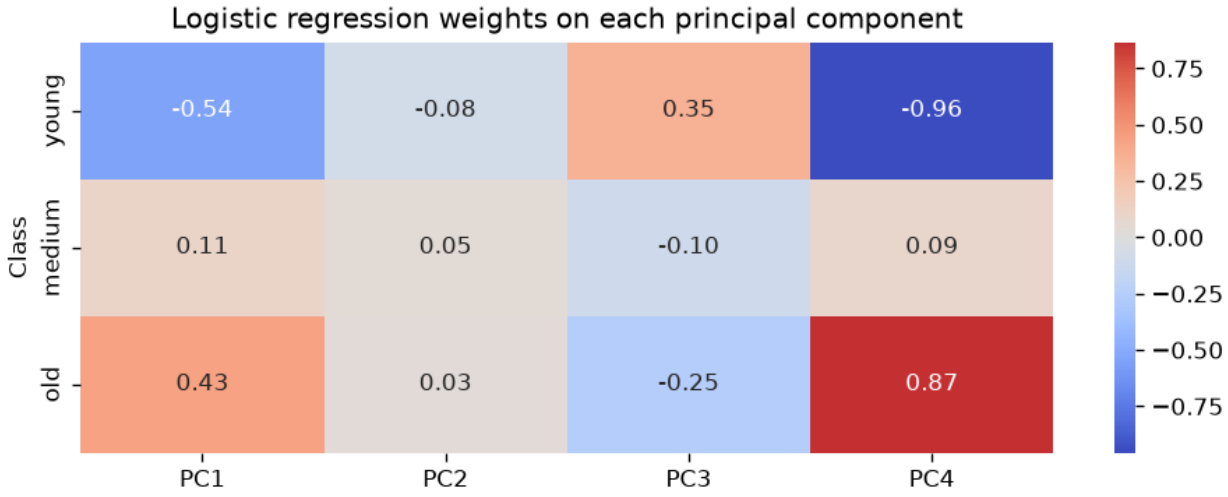
Component	Share	Cumulative
PC1	76.75%	76.75%
PC2	12.81%	89.57%
PC3	4.45%	94.02%
PC4	2.33%	96.35%

Length, diameter, and the weights are almost the same fact ('how big is it'), so PC1 alone holds 77% of the variance. The rule used here: keep the smallest number of components that still hold 95% of

the variance. That is 4 of 9. The other five are dropped before the second logistic regression.

The heatmap is logit_pca.coef_. Rows are classes, columns are PC1–PC4. A positive cell means that component pushes the prediction toward that class. Young has a negative PC1 weight (-0.54) and old has a positive one (+0.43): the big 'size' direction points from young toward old. Medium's weights are all small, which matches it being the class the model is unsure about.

Test accuracy on the 4 components falls to 59.69%. The dropped components still carried some age signal. PCA here buys a smaller input, and it costs about 6 percentage points.



PC1 separates young (negative) from old (positive). Medium barely uses any component.

Say this: Four components keep 96% of the variance. PC1 is size. Accuracy drops from 66% to 60% because some age signal lived in the components we dropped.

Part B. SVM, and the four-way comparison

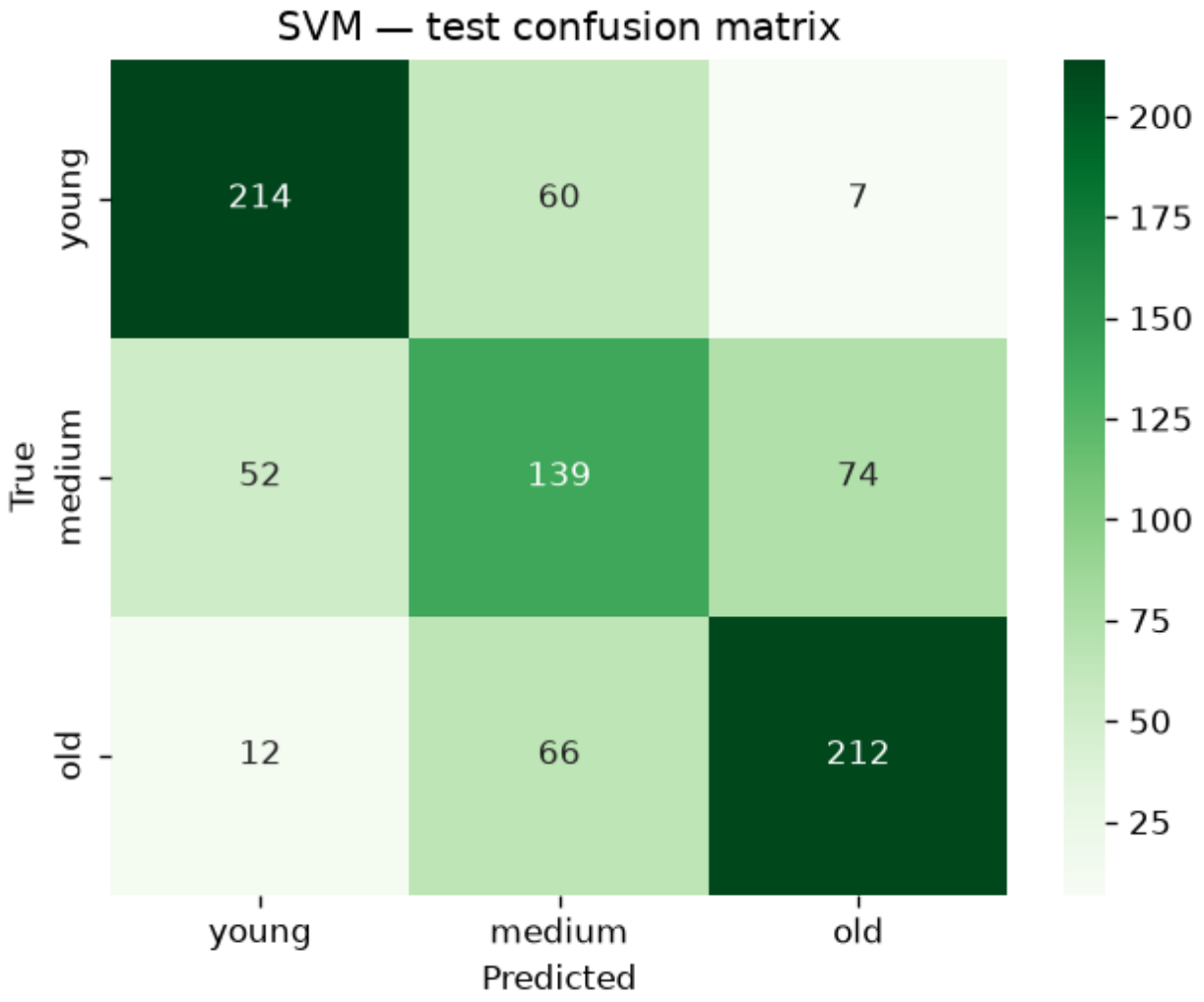
An SVM looks for the widest gap between classes. kernel='rbf' lets the boundary curve. C=1.0 is the penalty for points on the wrong side of that gap. gamma='scale' sets the width of the Gaussian from the data. The same training rows, the same scaler, and the same 4 principal components are reused, so the comparison is fair.

Model	Features	Test accuracy
Logistic regression	9 standardized columns	66.15%
Logistic regression	first 4 PCs	59.69%
SVM, RBF kernel	9 standardized columns	67.58%
SVM, RBF kernel	first 4 PCs	62.32%

The RBF SVM on the original standardized columns is the best of the four, at 67.58%. It only barely beats logistic regression (66.15%). The classes overlap in size, so a curved boundary helps a little, not a lot. PCA lowers both models by about 6 points. Use PCA here when you want fewer, uncorrelated inputs. Use the full standardized columns when you want the best test accuracy.



SVM on the full scaled features is the best bar. Both PCA models sit lower.



Same pattern as logistic regression: young and old are recognised, medium is the confused class.

Say this: Best test accuracy is the RBF SVM on the 9 scaled columns, 67.6%. PCA to 4 components keeps 96% of the variance and costs about 6 points of accuracy.